

Random walk in a null recurrent birth-and-death dynamical environment

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Abstract

We study a random walk in $\mathbb{Z}^d$ in a dynamical random environment constituted of independent null recurrent birth-and-death chains, and establish a scaling limit result for the position of the walker in terms of a sublinear time change of Brownian motion, thus establishing subdiffusivity, in contrast to a diffusivity result established previously in an ergodic version of the model.

AMS 2020 Subject Classifications: 60K37, 60F05

Keywords and Phrases: random walk in random environment, dynamical environment, birth-and-death environment, null recurrent environment, scaling limit, subdiffusivity

1 Introduction

This article may be seen as a report on a followup investigation on the continuous time random walk in dynamical random environment studied in [8]. As in that paper, we have an embedded (discrete time) mean zero random walk on $\mathbb{Z}^d$, which we presently take to be simple and symmetric for simplicity. The environment is also given by a family of independent and equally distributed birth-and-death processes indexed by $\mathbb{Z}^d$, but now, in contrast to [8], they are not ergodic; for simplicity, we consider spatially homogeneous transitions only (off the origin) for the environment. The jump rate of the random walk at space time point (x, t) is again given by a decreasing function of the environment at (x, t) , which we now take to have a specific form, with a specific decay, namely, (in the notation of [8]), $\varphi(x) = (x + 1)^{-\alpha}$, with $\alpha \in (0, 2)$ for null recurrent environments¹.

Our main result, Theorem 1, stated in the next section for the null recurrent case, is the scaling limit of the suitably rescaled position of the walker; the limit is a time changed Brownian Motion, with the time change a polynomial function of time, with degree less than one, amounting to a subdiffusive process, in contrast to the usual/diffusive CLT obtained in [8], under 'strong' ergodicity (according to the tentative terminology adopted in that paper). Our strategy requires that the dimension be at least five. There is no

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[†]Supported in part by CNPq grants 309938/2025-2 and 408590/2024-6, and FAPESP grant 2023/13453-5

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[§]Supported in part by FAPESP grants 2023/13453-5 and 2025/27064-6

¹which is the main case we study here, but we do discuss the (simpler) transient case of the environment, briefly at the end of the paper, where we have $\alpha \in (0, 1)$ instead

Theorem 1. Let $d \geq 5$, and, for $\delta > 0$, set $X^{(\delta)} = (X_t^{(\delta)})_{t \geq 0}$, where

$$X_t^{(\delta)} = \delta^{\frac{2-\alpha}{4}} X_{\delta^{-1}t}, \quad t \geq 0. \quad (2.3)$$

Then for a.e. realization of the environment

$$X^{(\delta)} \Rightarrow (B_{t^{1-\frac{\alpha}{2}}})_{t \geq 0} \text{ as } \delta \rightarrow 0, \quad (2.4)$$

in the finite dimensional sense, where $(B_t)_{t \geq 0}$ is Brownian motion with diffusion coefficient $1/\mu$, with $\mu \in (0, \infty)$ to be specified below — see (4.45).

Remark 2. The above convergence ought to hold in a stronger sense as well, namely, in Skorohod space, at least, conceivably, at high dimensions, but we avoid a stronger statement in order to not go too deep into technicalities involved in verifying standard criteria for such kinds of convergence, which, we expect, would mostly involve standard arguments.

3 Proof of Theorem 1

Our setup allows for a time change which makes the environment of "order 1", as follows. Let $W_0^x = 0$, $W_x(t) = (\eta_t^x + 1)/\sqrt{t}$, $t > 0$, $x \in \mathbb{Z}^d$, and set

$$Y_t = X_{t^\gamma}, \quad t \geq 0, \quad (3.1)$$

where $\gamma = \frac{2}{2-\alpha}$.

Lemma 3.1. Given H , $(Y_t)_{t \geq 0}$ is a Markov jump process starting at the origin, with jump chain $(\mathcal{X}_t)_{t \geq 0}$, and instantaneous jump rates at (x, t) given by $\rho_t^x := \gamma(W_{t^\gamma}^x)^{-\alpha}$, $t \geq 0$, $x \in \mathbb{Z}^d$.

Proof. This is a straightforward argument, which we outline for completeness.

That (Y_t) inherits the Markovian character and jump chain from (X_t) is quite clear. It remains to characterize the jump times of (Y_t) , which we will relate to those of (X_t) , as follows. The latter may be obtained, via a standard Harris graphical construction, from the family $\{\mathcal{P}^x, x \in \mathbb{Z}^d\}$ of independent Poisson point processes on $[0, \infty)$ with intensity function λ^x , respectively.

It is now enough to check that for each $x \in \mathbb{Z}^d$, $\mathcal{P}^x$ is mapped, by the transformation $t \rightarrow t^\gamma$, to $\tilde{\mathcal{P}}^x$, a Poisson point processes on $[0, \infty)$ with intensity function $\gamma(W_{t^\gamma}^x)^{-\alpha}$, $t \geq 0$.

That such a transformation maps a Poisson point process to a Poisson point process is quite clear, so it is enough to obtain the expected value of the occupation random variables of $\tilde{\mathcal{P}}^x$, namely, $\mathbb{E}(\#(r, s))$, $0 \leq r < s < \infty$, where

$$\#(r, s) = |[r, s] \cap \tilde{\mathcal{P}}^x|, \quad (3.2)$$

where $|\cdot|$ means cardinality of $\cdot$.

But the right hand side of (3.2) equals $|[r^\gamma, s^\gamma] \cap \mathcal{P}^x|$, and

$$\mathbb{E}\{|[r^\gamma, s^\gamma] \cap \mathcal{P}^x|\} = \int_{r^\gamma}^{s^\gamma} \lambda_t^x dt. \quad (3.3)$$

We conclude by resorting to an obvious change of variables in the integral to find that it equals $\int_r^s \rho_t^x dt$. $\square$

4.2 Sublinear bound on successive times between first and last visits to new sites

For $n \geq 0$, let $\hat{\omega}_n$ the (end) time of the last visit by (Y_t) at $\mathcal{Y}_n$, that is,

$$\hat{\tau}_n = \sup\{t \geq 0 : Y_t = \mathcal{Y}_n\}. \quad (4.22)$$

Lemma 4.2.1. *Let $\hat{\kappa}_n = \hat{\omega}_n - \hat{\tau}_n$, $n \geq 0$. If $d \geq 5$, we may choose a constant $\delta' < 1$ such that*

$$\sum_{n \geq 0} \mathbb{P}(\hat{\kappa}_n > \hat{\tau}_n^{\delta'}) < \infty. \quad (4.23)$$

Proof. We write

$$\hat{\kappa}_n = \sum_{j=1}^{\kappa_n} \mathcal{E}_j^n, \quad (4.24)$$

where $\mathcal{E}_1^n$ is the duration of the visit by (Y_t) to $\mathcal{X}_{\tau_n}$ starting at $\mathcal{T}_0^n := \hat{\tau}_n$, and, inductively, for $j \geq 2$, $\mathcal{E}_j^n$ is the duration of the visit by (Y_t) to $\mathcal{X}_{\tau_n+j-1}$ starting at $\mathcal{T}_{j-1}^n := \hat{\tau}_n + \sum_{i=1}^{j-1} \mathcal{E}_i^n$.

By (4.7), we may assume that $\kappa_n \leq n^\delta$ for $\delta \in (\frac{2}{d-2}, 1)$ fixed. We may also assume, by Lemma 4.1.2, that $\tau_n > an$.

Let us fix $c \in (0, 1 - \delta)$. By Markovianess and spatial homogeneity,

$$\begin{aligned} \mathbb{P}(\mathcal{E}_1^n > t_0^c | \hat{\tau}_n = \mathcal{T}_0^n = t_0, Y_{t_0} = y) &= \mathbb{E} \left[\exp \left\{ -\gamma \int_{t_0}^{t_0+t_0^c} (W_{s^\gamma}^0)^{-\alpha} ds \right\} \right] \\ &\leq \mathbb{E} (e^{-\gamma t_0^c (M_{t_0})^{-\alpha}}), \end{aligned} \quad (4.25)$$

where $M_t = \max \{W_s^0 : t^\gamma \leq s \leq (t + t_0^{c+\delta})^\gamma\}$, $t \geq 0$.

Let us set $c' = c/(2 + \alpha)$. Given $t_0 > an$, we have that

$$\begin{aligned} \mathbb{P}(M_{t_0} > t_0^{c'}) &\leq \mathbb{P} \left(\max_{t_0^\gamma \leq s \leq (t_0+t_0^{c+\delta})^\gamma} |S_{N_s^0}^0| > t_0^{\gamma/2} t_0^{c'} - 1 \right) \\ &= \mathbb{P} \left(\max_{\substack{N_0^0 \leq k \leq N_{(t_0+t_0^{c+\delta})^\gamma}^0}} |S_k^0| > t_0^{\gamma/2} t_0^{c'} - 1 \right). \end{aligned} \quad (4.26)$$

Resorting to standard large deviation bounds for (N^0) , it is enough to bound

$$\mathbb{P} \left(\max_{\frac{1}{2}t_0^\gamma \leq k \leq \frac{3}{2}t_0^\gamma} |S_k^0| > t_0^{\gamma/2} t_0^{c'} - 1 \right) \leq e^{-c'' t_0^{2c'}}, \quad (4.27)$$

where $c'' > 0$ is a constant, and the latter inequality follows from the union bound and Bernstein Inequality. It follows that

$$\mathbb{P}(\mathcal{E}_1^n > t_0^c | \mathcal{T}_0^n = t_0, Y_{t_0} = y) \leq e^{-\gamma t_0^{c-\alpha c'}} + e^{-c'' t_0^{2c'}} \leq 2e^{-\gamma'' t_0^{2c'}}, \quad (4.28)$$

where $\gamma'' = \gamma \wedge c''$.

Proceeding inductively, we may repeat the latter approach to find that, for $1 \leq j < [n^\delta]$, we have that, for $t_0 \leq t_j \leq t_0 + jt_0^c$,

$$\begin{aligned} \mathbb{P}(\mathcal{E}_{j+1}^n > t_0^c | \hat{\tau}_n = t_0, \mathcal{T}_j^n = t_j, Y_{t_j} = y) &= \mathbb{E} \left[\exp \left\{ -\gamma \int_{t_j}^{t_j+t_0^c} (W_{s^\gamma}^0)^{-\alpha} ds \right\} \right] \\ &\leq \mathbb{E} (e^{-\gamma t_0^c (M_{t_j})^{-\alpha}}), \end{aligned} \quad (4.29)$$

and the bound on the right hand side of (4.28) follows from the left hand side of (4.29) with a similar argument as above.

It follows that outside of an event of probability bounded above by $2n^\delta e^{-c''n^{2ac'}}$ (on account of the above estimates on $\mathcal{E}_j^n$, $1 \leq j \leq n^\delta$), combined with the estimates of the probability of $\{\kappa_n > n^\delta\}$, from (4.7), and of $\{\tau_n \leq an\}$, from Lemma 4.1.2, we have that, outside of an event with summable probability,

$$\hat{\kappa}_n \leq n^{\delta+c}, \quad (4.30)$$

and the result follows, using again Lemma 4.1.2, for any choice $\delta' \in (\delta + c, 1)$. $\square$

4.3 Conclusion of proof of Proposition 3

We will fix $(\mathcal{X}_n)$ and resort to a convenient construction of $\{(\eta_t^{\mathcal{Y}_n}), n \geq 0\}$, which is the relevant environment, as follows. By Skorohod representation, we may write $(S_\ell^{\mathcal{Y}_n})$ for each $n \geq 0$ as

$$S_\ell^{\mathcal{Y}_n} = \mathcal{B}_n(T_\ell^n), \ell \geq 0, \quad (4.31)$$

where $\mathcal{B}_n$, $n \geq 0$, are independent standard Brownian motions, and (T_ℓ^n) , $n \geq 0$, are iid sequences of iid non negative random variables with mean 1 and a positive exponential moment.

It is also quite clear that, for $n \geq 0$, the relevant environment in $\mathcal{Y}_n$ is the one between times $\hat{\tau}_n^\gamma$ and $(\hat{\tau}_n + \hat{\kappa}_n)^\gamma$, namely

$$(\eta_t^{\mathcal{Y}_n})_{\hat{\tau}_n^\gamma}^{(\hat{\tau}_n + \hat{\kappa}_n)^\gamma} = \left(\frac{|\mathcal{B}_n(T_{N_t^n}^n)| + 1}{\sqrt{t}} \right)_{\hat{\tau}_n^\gamma}^{(\hat{\tau}_n + \hat{\kappa}_n)^\gamma} = \left(\frac{|\mathcal{B}_n(T_{N_t^n}^n)|}{\sqrt{t}} + \frac{1}{\sqrt{t}} \right)_{\hat{\tau}_n^\gamma}^{(\hat{\tau}_n + \hat{\kappa}_n)^\gamma}, \quad (4.32)$$

where $(N_t^n) = (N_t^{\mathcal{Y}_n})$, $n \geq 0$. We now write the the first term within parentheses on the right hand side, for $n \geq 1$, as

$$\left| \frac{\mathcal{B}_n(T_{N_t^n}^n)}{\hat{\tau}_n^{\gamma/2}} \right| \times \sqrt{\frac{\hat{\tau}_n^\gamma}{t}} = |\tilde{\mathcal{B}}_n(\tilde{T}_t^n)| \times \sqrt{\frac{\hat{\tau}_n^\gamma}{t}}, \quad (4.33)$$

where $\tilde{T}_t^n = T_{N_t^n}^n / \hat{\tau}_n^\gamma$, $t \geq \hat{\tau}_n^\gamma$, and $\tilde{\mathcal{B}}_n$, $n \geq 1$, are independent standard Brownian motions. The latter equality follows from well known scale invariance of Brownian motion, once we observe that $\hat{\tau}_n$ and $\mathcal{B}_n$ are independent.

Now, for $n \geq 0$, let

$$\begin{aligned} W_n^- &= \min \{W_t^{\mathcal{Y}_n} : \hat{\tau}_n^\gamma \leq t \leq (\hat{\tau}_n + \hat{\kappa}_n)^\gamma\}, \\ W_n^+ &= \max \{W_t^{\mathcal{Y}_n} : \hat{\tau}_n^\gamma \leq t \leq (\hat{\tau}_n + \hat{\kappa}_n)^\gamma\}; \end{aligned}$$

and, given $\varepsilon \in (0, 1)$, set

$$\begin{aligned} V_n^- &= V_n^-(\varepsilon) = (1 - \varepsilon) \min \{|\tilde{\mathcal{B}}_n(t)| : 1 - \varepsilon \leq t \leq 1 + \varepsilon\}, \\ V_n^+ &= V_n^+(\varepsilon) = \varepsilon + \max \{|\tilde{\mathcal{B}}_n(t)| : 1 - \varepsilon \leq t \leq 1 + \varepsilon\}. \end{aligned}$$

From standard large deviation estimates for Poisson processes and sums of iid random variables with a positive exponential moments, as well as Lemmas 4.1.2 and 4.2.1 above, we have that, except for an event whose probability is summable in n ,

$$V_n^- \leq W_n^- \leq W_n^+ \leq V_n^+, \quad (4.34)$$